

KAIROUAN, Tunisia

WMO: 607350

Lat: 35.667N

Lon: 10.100E

Elev: 223

StdP: 14.58

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.9	43.0	27.4	21.5	55.0	30.0	24.3	54.9	17.2	58.3	14.3	57.8	2.4	220	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.6	107.5	71.6	103.7	71.2	100.3	70.9	77.4	90.3	76.1	89.3	74.8	88.6	4.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.3	129.4	82.1	72.7	122.3	81.6	71.2	116.2	80.9	40.9	90.4	39.6	89.3	38.4	88.5	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	11.9	10.1	36.3	113.8	2.3	2.9	34.6	115.9	33.3	117.6	32.0	119.2	30.3	121.3		
(5)			33.6	80.5	2.0	2.4	32.2	82.3	31.0	83.7	29.9	85.0	28.5	86.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.1	54.8	55.8	60.6	65.7	73.4	81.6	87.3	87.1	80.1	72.9	63.4	57.0		
	DBStd	12.53	4.13	4.63	5.50	4.91	5.33	5.59	4.18	4.43	5.21	5.10	4.66	4.26		
	HDD50	17	7	6	1	0	0	0	0	0	0	0	0	3		
	HDD65	1123	316	261	159	49	2	0	0	0	0	3	82	251		
	CDD50	7340	158	168	329	472	726	947	1157	1149	902	710	403	219		
	CDD65	2968	1	2	21	71	263	497	692	684	452	248	35	2		
	CDH74	33789	11	34	264	638	2417	5761	9270	9009	4455	1722	193	15		
	CDH80	18378	0	4	69	180	1060	3277	5681	5369	2108	600	29	1		
(14) Wind	WSAvg	4.0	3.6	4.1	4.3	4.7	4.7	4.4	4.5	3.8	3.6	3.2	3.4	3.5		
(15) Precipitation	PrecAvg	11.5	0.9	0.9	1.3	1.0	0.9	0.4	0.1	0.5	1.7	1.7	1.0	1.0		
	PrecMax	28.9	4.8	4.3	3.7	2.6	3.5	2.1	1.3	4.0	7.5	16.9	4.1	5.0		
	PrecMin	5.3	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0		
	PrecStd	4.2	1.0	0.9	0.9	0.7	0.8	0.5	0.2	0.7	1.6	2.4	1.0	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.1	78.8	87.5	90.0	101.9	111.4	112.4	111.6	103.7	95.3	84.0	75.6		
		MCWB	56.0	58.5	60.8	62.6	69.2	69.7	71.7	72.7	71.2	68.9	63.2	57.5		
	2%	DB	69.9	73.2	80.4	84.8	94.7	103.5	107.3	107.2	98.2	89.9	78.9	71.4		
		MCWB	54.6	56.3	59.1	62.0	66.6	69.9	71.7	72.5	72.2	68.5	62.4	56.6		
	5%	DB	66.2	69.2	76.0	80.8	89.8	98.7	103.6	103.0	94.6	86.2	75.5	68.1		
		MCWB	53.3	54.5	57.9	60.7	65.7	68.9	71.1	72.2	72.4	68.4	61.2	55.4		
	10%	DB	63.2	65.8	71.8	77.2	85.8	94.9	100.1	99.3	90.9	82.6	72.7	65.8		
		MCWB	52.4	53.2	56.6	59.8	64.4	68.7	70.7	72.0	71.8	67.6	59.9	54.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.3	60.0	63.9	66.4	71.7	75.4	77.8	80.2	78.8	74.7	68.0	61.2		
		MCDB	67.1	73.6	77.7	81.0	92.1	92.2	92.7	96.0	90.5	83.4	77.7	69.4		
	2%	WB	56.4	57.5	61.3	63.9	69.2	73.0	76.1	78.0	76.8	72.7	65.5	58.8		
		MCDB	65.2	69.1	74.8	77.3	87.2	90.8	90.9	91.7	87.1	80.6	72.9	66.1		
	5%	WB	55.0	55.8	59.5	62.6	67.5	71.6	74.8	76.4	75.3	71.1	63.5	57.2		
		MCDB	63.1	66.0	71.6	75.9	83.8	88.4	90.4	90.1	86.3	79.6	71.3	65.6		
	10%	WB	53.6	54.3	57.9	61.2	66.0	70.3	73.5	75.2	73.8	69.6	61.7	55.8		
		MCDB	61.8	63.7	68.7	73.0	81.2	87.7	90.3	89.2	85.2	78.8	70.0	63.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	18.2	20.2	20.8	23.0	24.9	25.6	24.1	19.7	18.3	17.6	16.8		
		MCDBR	22.4	24.8	27.5	27.8	29.6	30.3	30.2	29.2	24.6	23.3	21.8	21.1		
		MCWBR	11.5	12.0	11.1	10.2	9.1	8.3	8.3	8.2	7.6	8.3	9.9	10.9		
	5% WB	MCDBR	18.9	21.7	24.0	23.7	26.1	26.8	26.4	24.9	21.6	19.4	18.1	17.8		
		MCWBR	10.6	11.2	10.6	9.4	9.0	8.9	8.4	8.1	8.1	7.7	9.5	10.3		
(40) Clear-Sky Solar Irradiance	taub		0.337	0.364	0.431	0.481	0.503	0.473	0.498	0.489	0.483	0.439	0.384	0.347		
	taud		2.379	2.292	2.084	1.973	1.943	2.056	2.001	2.024	2.050	2.130	2.254	2.358		
	Ebn at Noon		270	275	264	257	252	259	252	252	245	245	248	258		
	Edh at Noon		30	36	48	57	60	53	56	53	49	42	33	29		
(44) All-Sky Solar Radiation	RadAvg		874	1129	1510	1854	2177	2425	2462	2176	1661	1281	945	777		
	RadStd		58	70	84	115	109	102	49	83	72	68	43	54		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.70	N/A	N/A	N/A	N/A	N/A	N/A	-131	+256	N/A
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page